CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Stormwater Rescue

Goal: Model stormwater runoff and erosion, then test one intervention that slows, redirects, absorbs, or filters the flow.

35 minutes | 8 steps | Engineering and Design | Playbook B meeting 25

Ask | Plan | Build | Test | Improve

You are done when

Compare baseline and revised evidence, then state what the intervention changed and what the model cannot prove.

You will use

- Sloped trays
- Model soil or sand
- Measured water cups
- Collection cups
- Gravel
- Sponge
- Craft sticks
- Mesh
- Model plants
- Towels
- Scissors
- Wet mats
- Return tub

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Place the sloped tray on the wet mat.
- Measure the rain water.

2 Step 2: Stormwater safety — 4 min

- Keep wet material inside the tray.
- Wipe each spill with a towel.
- Cut mesh only on the table.
- Keep fingers outside the cutting path.
- Taste no test material.

3 Step 3: Run the first test — 5 min

- Pour the measured rain from the marked height.
- Measure the water that leaves the tray.
- Mark where soil moved.

4 Step 4: Build a rescue — 6 min

- Choose one problem location.
- Cut only the planned mesh piece.
- Build one helper that slows or moves the water.

5 Step 5: Repeat the rain — 6 min

- Pour the same water amount.
- Measure the water that leaves the tray.
- Mark where soil moved this time.

6 Step 6: Improve the rescue — 3 min

- Change one part of the helper.
- Predict the next result.

7 Step 7: Clean the tray — 5 min

- One table at a time, carry wet soil to the return tub.
- Place wet tools on the drying mat.
- Dry the tray and table.
- Wash hands when the teacher calls your table.

8 Step 8: Exit check — 3 min

- Compare both water amounts.
- Name the helpful feature.

What to turn in

Baseline and revised results, and what your change did.

No computer? Another way to do it

Watch one teacher demo and predict with drawings.

Finished early? Next challenge

Storm Upgrade: Predict how the intervention would perform with twice the water and identify the first likely failure.

Stay safe

Never drink test water, keep wet materials inside trays, wipe spills immediately, and wash hands after handling model soil.

Grade targets

Grade 5: Compare before and after with a measuring cup.
Grade 6: Explain what the model cannot prove.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.